

# Compressing the collective knowledge of ESM into a single protein language model

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# Motivation: Variant Effect Prediction (VEP)

- Goal: Predict functional impact of mutations
- VEP is central to understanding disease and protein function
- PLMs (like ESM) learn evolutionary constraints directly from sequences
- But best-performing methods today are **hybrid models**

# Limitations of Hybrid Approaches

- Hybrid methods use extra modalities:
  - MSA (TransceptEVE)
  - 3D structure (SAProt)
  - population data (AlphaMissense,)
- Issues:
  - High complexity/ computational cost
  - Limited coverage (structure unavailable)
  - Data dependency / bias (e.g potential circularity (especially with population data))
- Question:

*Are sequence-only PLMs fundamentally limited?*

# Key idea

- Are sequence-only PLMs are indeed **inherently limited?**
- Hidden evolutionary signals already exist in PLMs
- Problem = **signal under-utilization**
- Hypothesis:  
Different PLMs capture **different parts of evolutionary landscape**

# ESM Model Family Complementarity

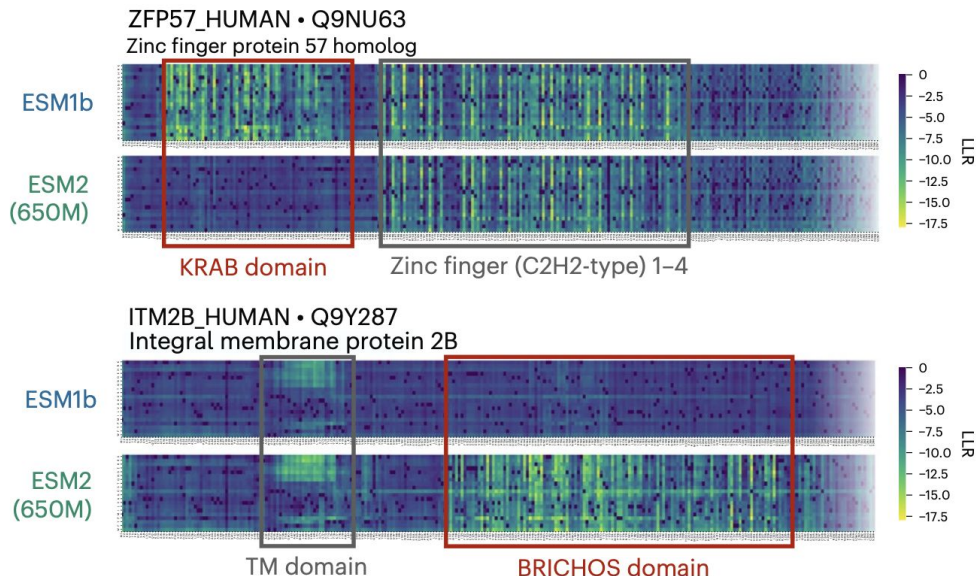
Different ESM models detect **different domains**

Example:

- ESM1b → KRAB domains
- ESM2 → BRICHOS domains

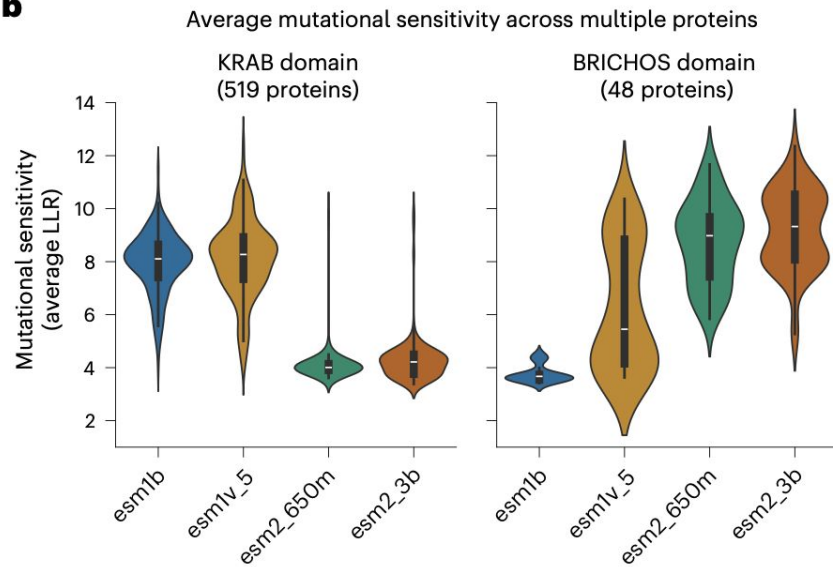
Same architecture → different blind spots

Evolutionary signal is **distributed across models**



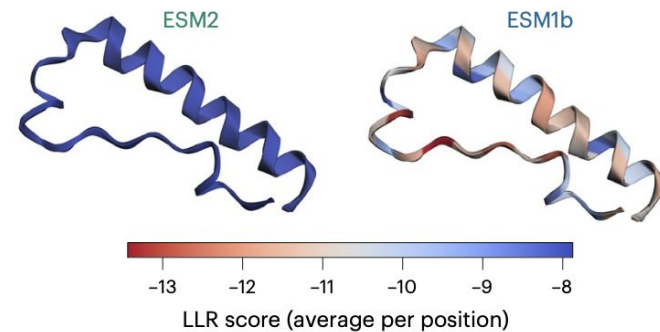
ESM LLRs on experimentally determined  
3D structures of KRAB and BRICHOS domains

**b**



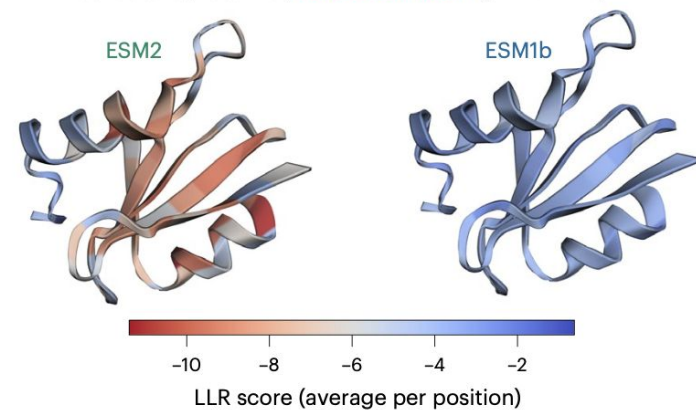
ZNF93\_HUMAN

Zinc finger protein 93 | KRAB domain (PDB: 7Z36)



PSPC\_HUMAN

Surfactant protein C | BRICHOS domain (PDB: 2YAD)



# Maximum Confidence Strategy (ESMIN)

For each mutation:

## Minimum Log Likelihood Ratio of mutation

Instead of averaging predictions,  
use **max confidence selection**

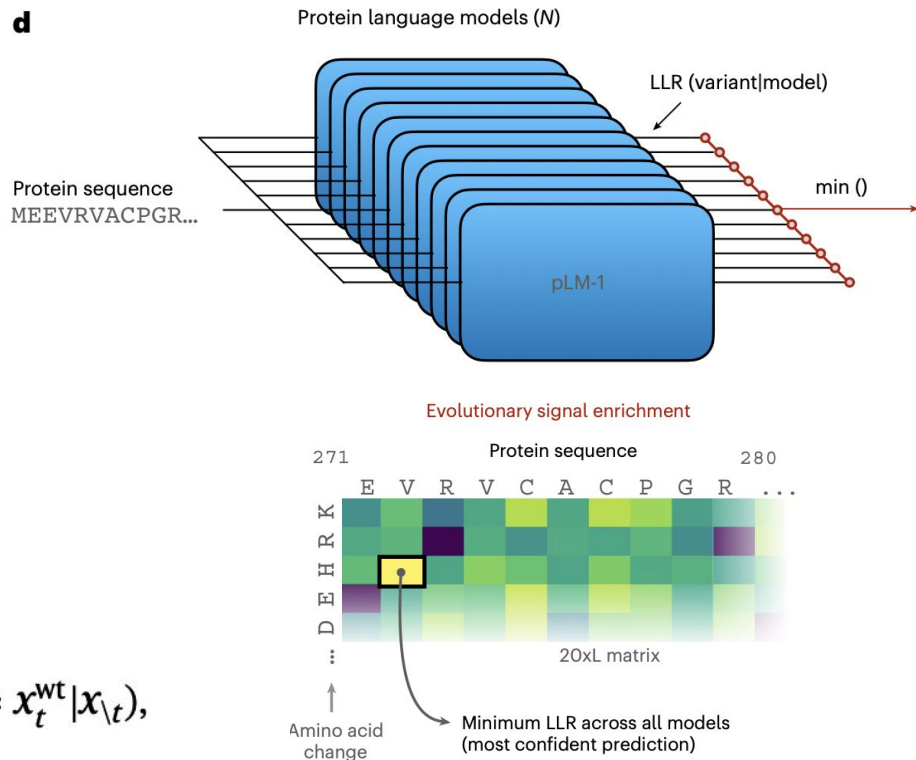
Intuition:

If *any* model strongly detects signal → trust it

Avoids:

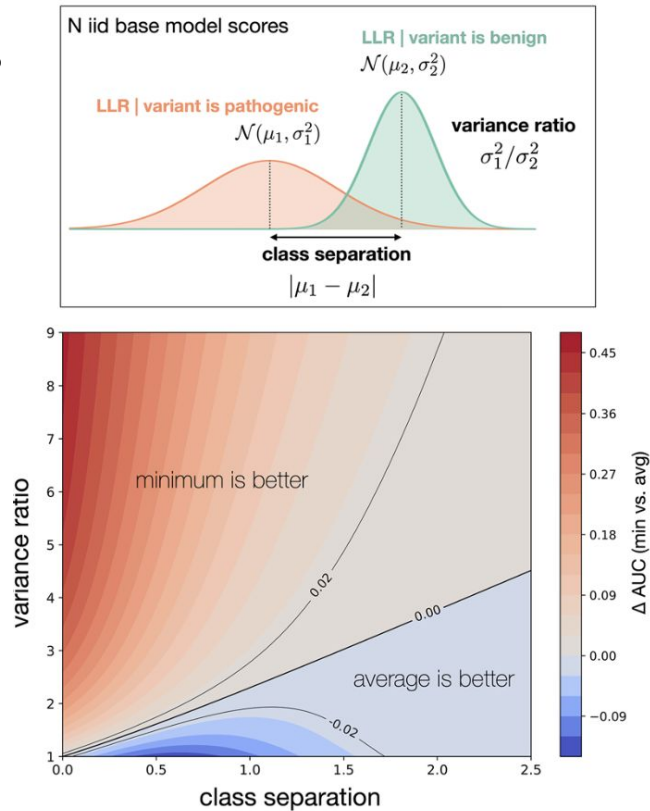
signal dilution from averaging

$$\text{LLR}(x^{\text{mt}}|x^{\text{wt}}) = \sum_{t \in T} \log p(x_t = x_t^{\text{mt}}|x_{\setminus t}) - \log p(x_t = x_t^{\text{wt}}|x_{\setminus t}),$$

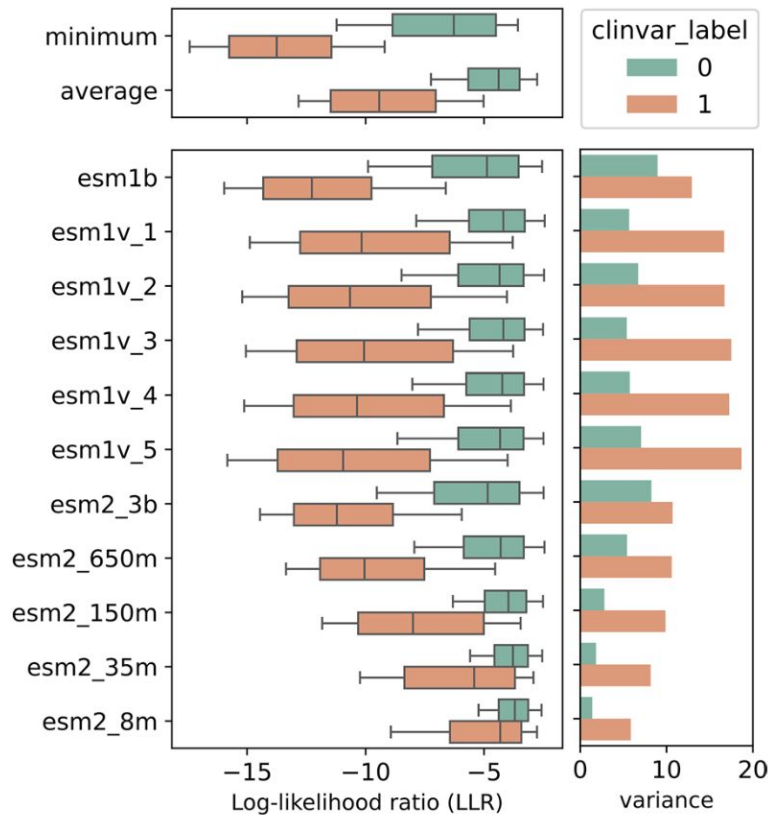


# Why Minimum Works

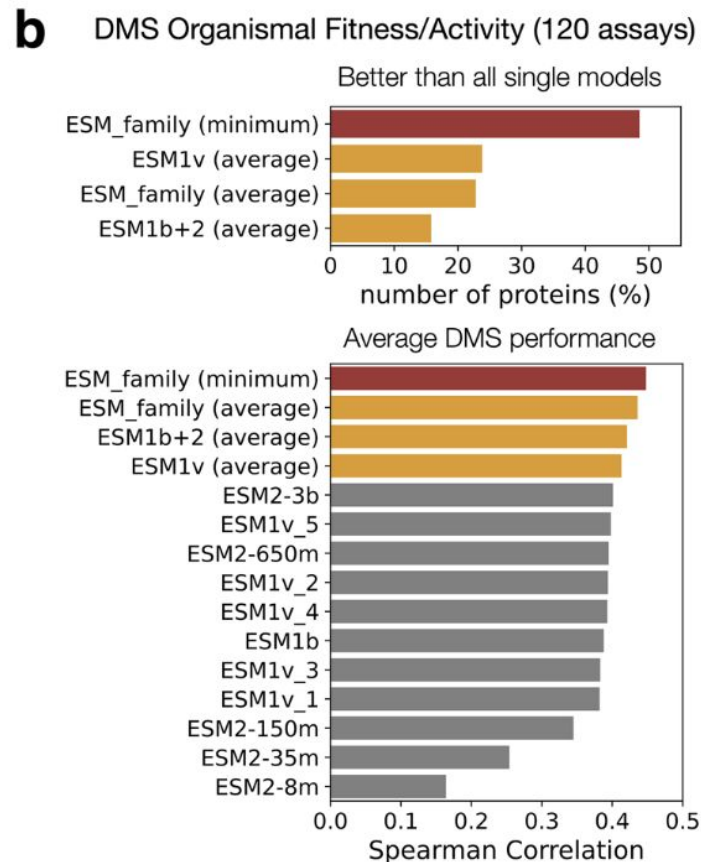
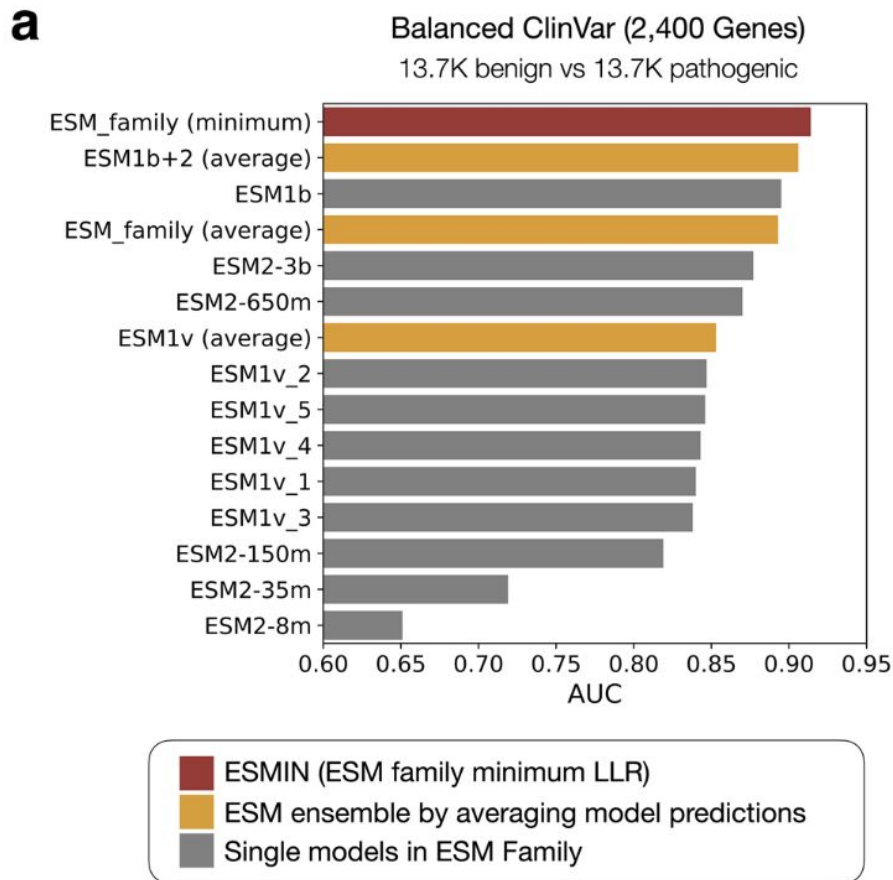
Theoretical comparison between minimum and average score aggregation



**b** ESM family LLR score distributions (base models, ESMIN and average ensembles)

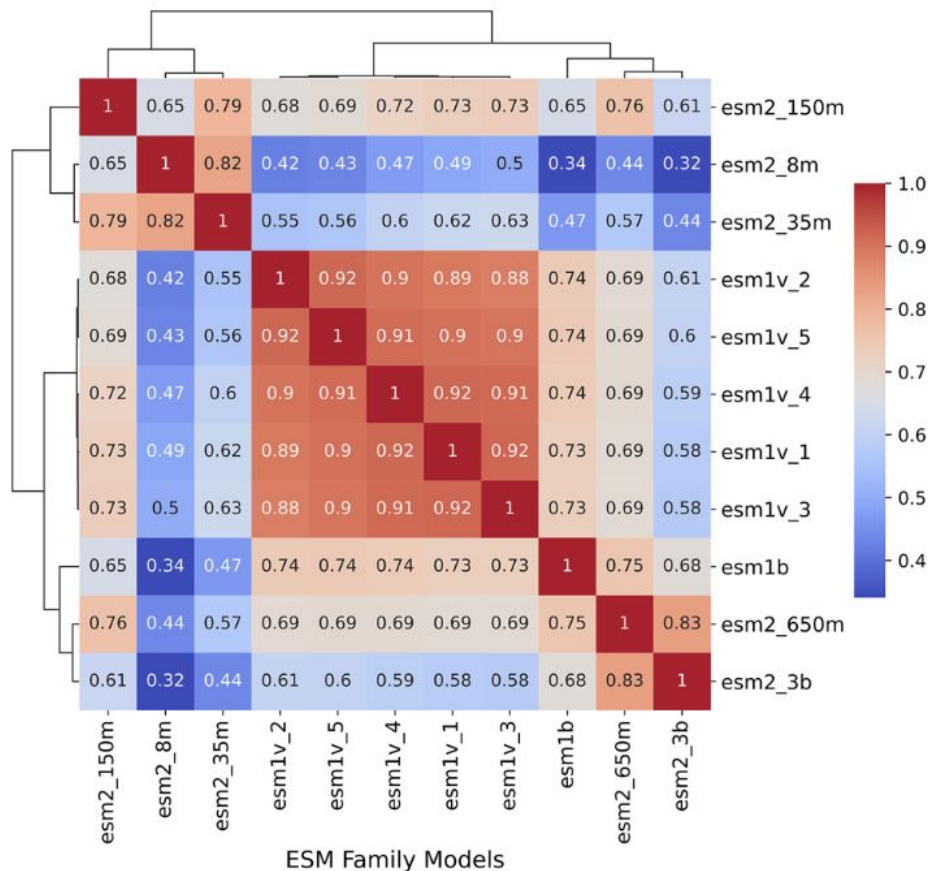


# ESMIN outperforms single sequence protein LM baselines



# Model Correlation and Composition

**c** Spearman Correlation (ESM Family Models)



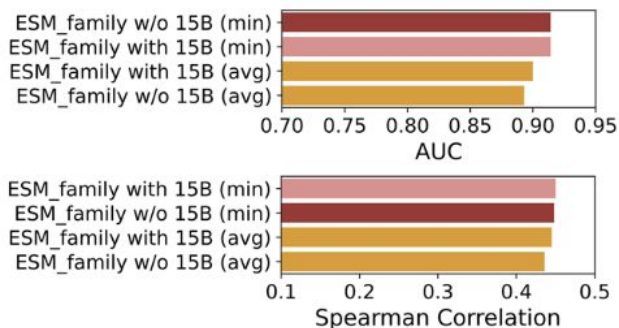
**d**

ESM Family minimum LLR (ESMIN)  
Model Composition and Correlation with ESMIN

| Model     | Fraction | Correlation |
|-----------|----------|-------------|
| esm1b     | 0.257    | 0.860       |
| esm2.3b   | 0.326    | 0.845       |
| esm2.650m | 0.050    | 0.771       |
| esm1v_2   | 0.085    | 0.762       |
| esm1v_5   | 0.101    | 0.755       |
| esm1v_4   | 0.050    | 0.729       |
| esm1v_1   | 0.041    | 0.711       |
| esm1v_3   | 0.040    | 0.706       |
| esm2_150m | 0.015    | 0.588       |
| esm2_35m  | 0.017    | 0.412       |
| esm2_8m   | 0.018    | 0.289       |

**e**

ESM Family with or without ESM2 (15B)



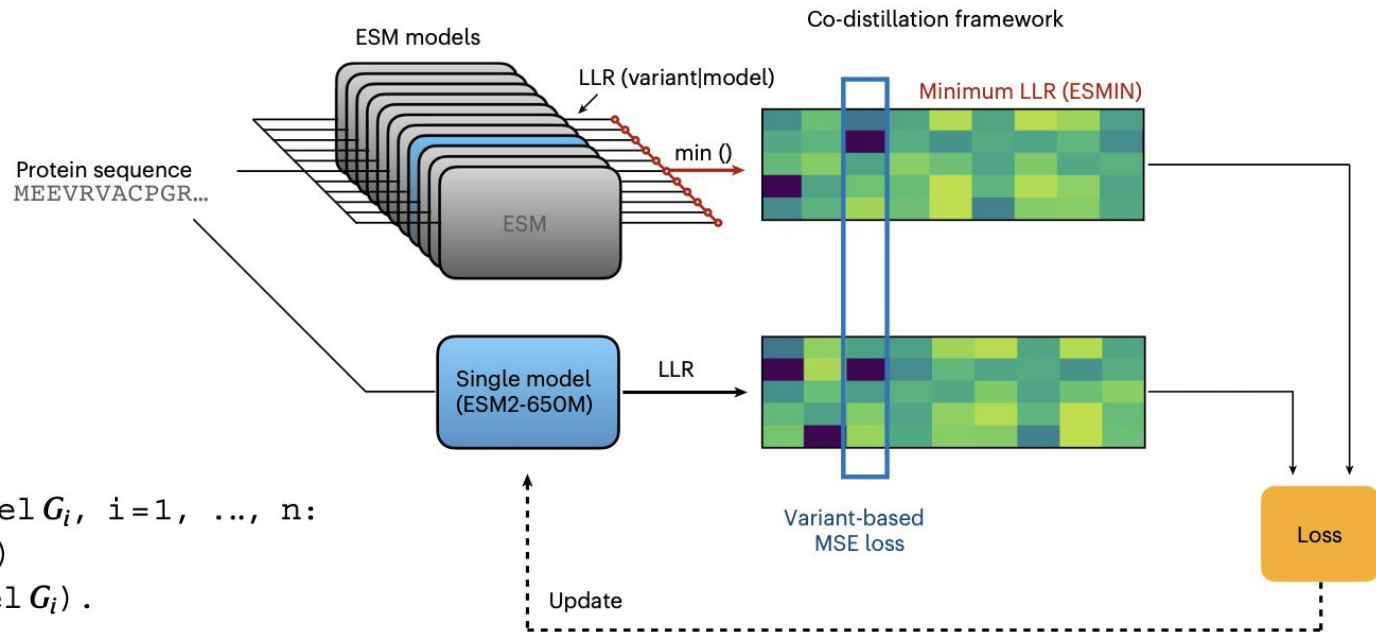
# Codistillation framework

Models teach each other

Use most confident prediction as supervisory

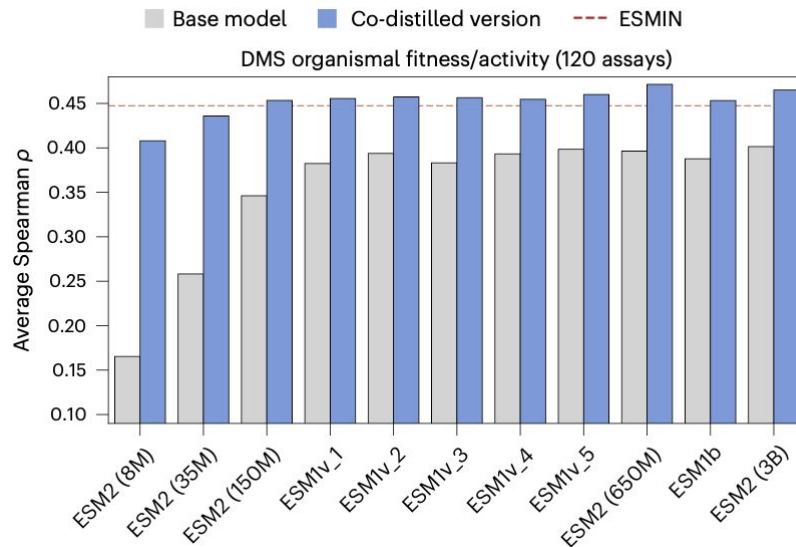
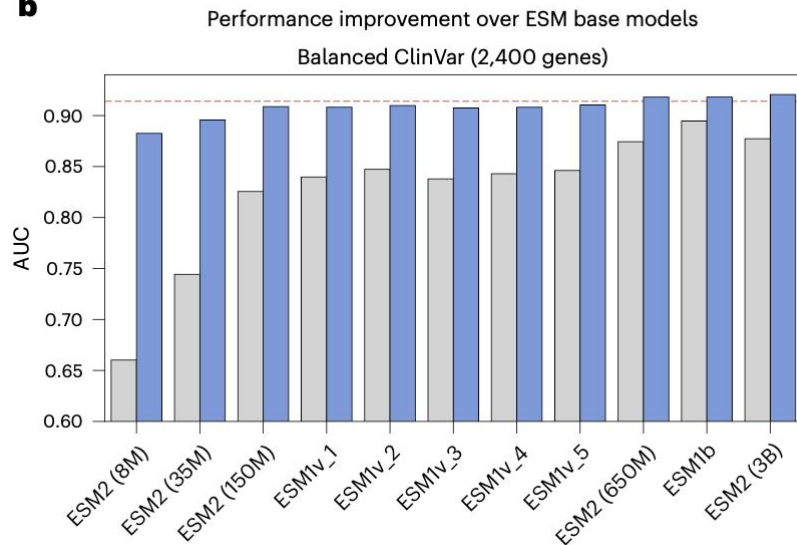
Iterative learning:  
teacher  $\leftrightarrow$  student

For variant  $v$  in  $V$  and model  $G_i$ ,  $i=1, \dots, n$ :  
 $\text{Loss}_i = \text{MSE}(F(v|G_i), \min F(v|G_s))$   
 $\text{backpropagate}(\text{loss}_i, \text{model } G_i)$ .



# Co-distilled models vs baselines on ClinVar and ProteinGym

**b**



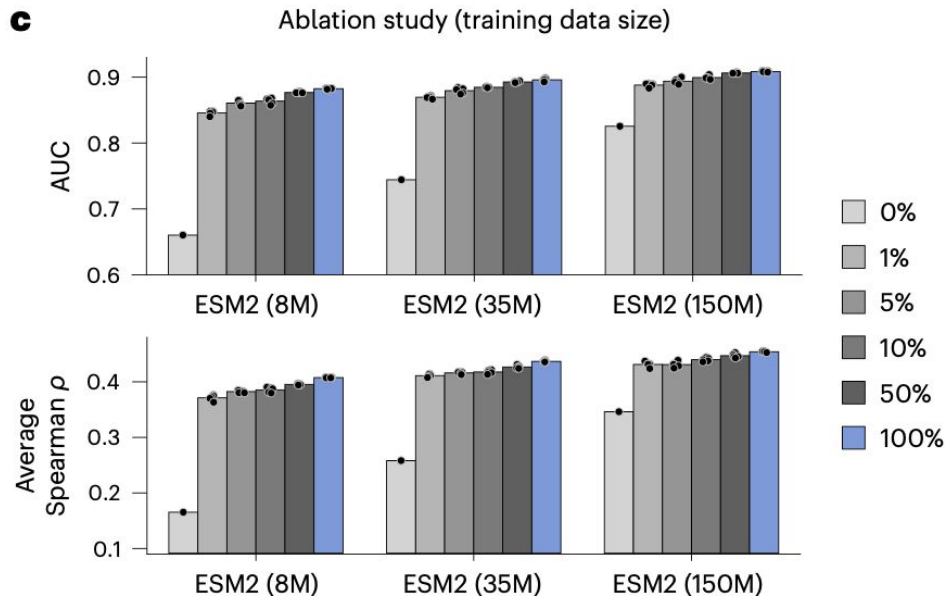
# Co-distillation is effective even with limited data

100% - 18,683 human proteins.

Exclude 5,279 proteins, (>30% sim), sample

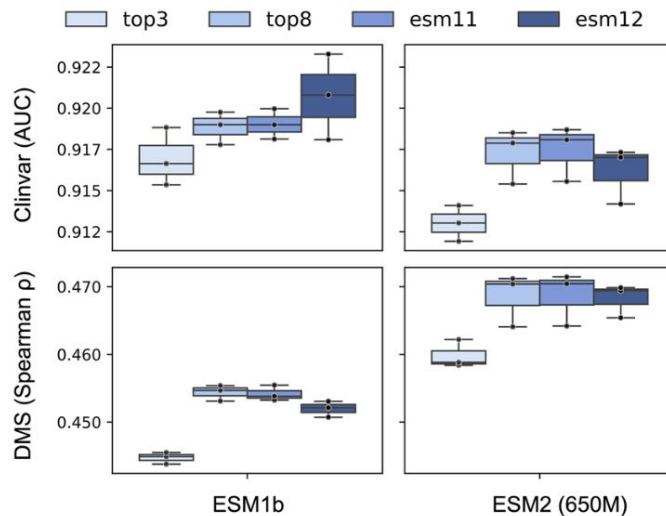
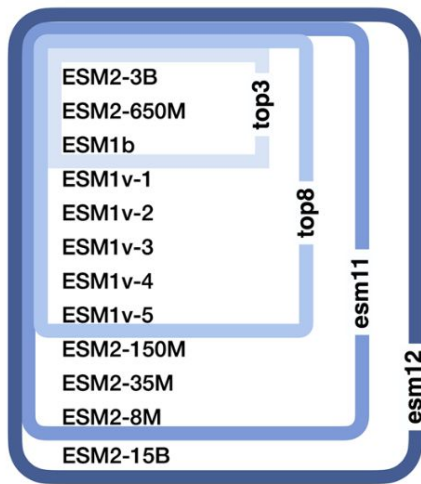
- 1%
- 5%
- 10%
- 50%

Zero shot: 0%

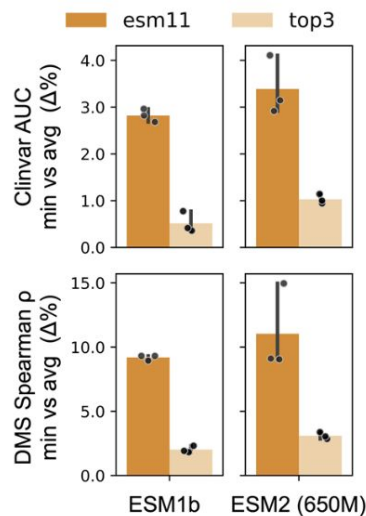


# How the choice of models/aggregation affects co-distillation

ESMIN model composition ablation (max-confidence co-distillation)



**d** min vs. avg aggregation



# Co-distilled model embeddings yield substantial gains on all variant-level tasks, without hurting performance protein-level tasks

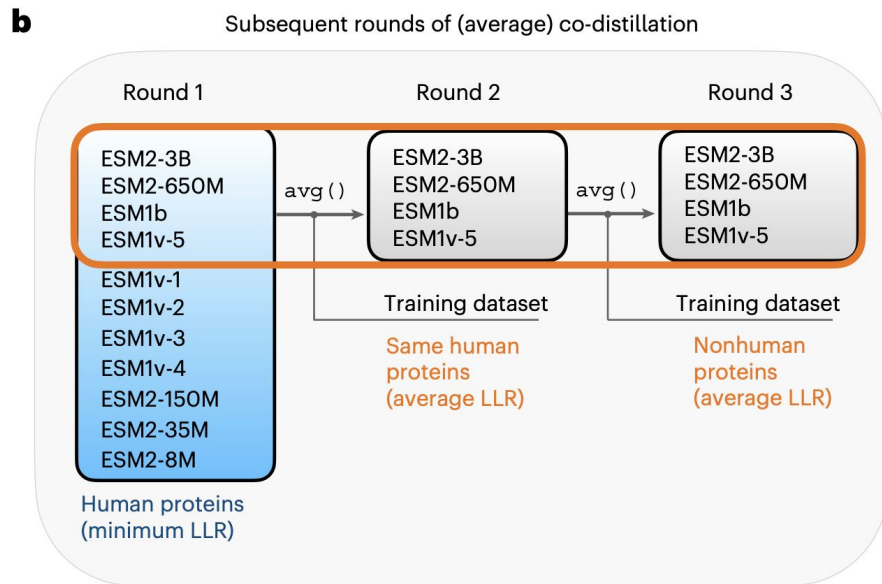
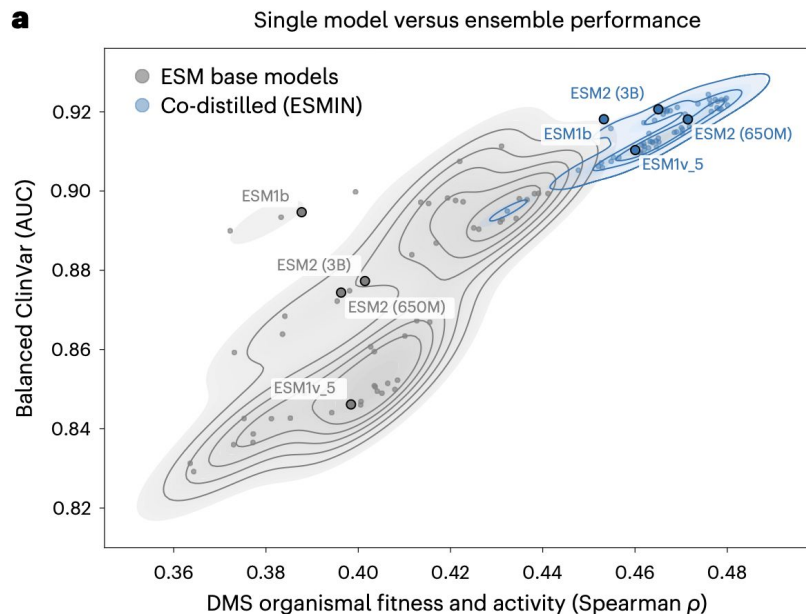
**d**

Fine-tuning using model embeddings  
(variant-level and protein-level downstream tasks)

| Task               | Metric   | ESM2-650M<br>(base) | ESM2-650M<br>(co-distilled) |
|--------------------|----------|---------------------|-----------------------------|
| AAV                | Spearman | $0.465 \pm 0.031$   | $0.573 \pm 0.022$           |
| Fluorescence       | Spearman | $0.560 \pm 0.005$   | $0.580 \pm 0.007$           |
| $\beta$ -Lactamase | Spearman | $0.686 \pm 0.014$   | $0.740 \pm 0.016$           |
| Stability          | Spearman | $0.714 \pm 0.011$   | $0.767 \pm 0.012$           |
| HumanPPI           | Accuracy | $0.874 \pm 0.003$   | $0.887 \pm 0.006$           |
| DeepLoc-2          | Accuracy | $0.909 \pm 0.005$   | $0.900 \pm 0.004$           |
| DeepLoc-10         | Accuracy | $0.800 \pm 0.008$   | $0.795 \pm 0.003$           |
| MetallionBinding   | Accuracy | $0.684 \pm 0.016$   | $0.681 \pm 0.009$           |
| Thermostability    | Spearman | $0.695 \pm 0.003$   | $0.706 \pm 0.001$           |

# Iterative co-distillation converges to a single better model

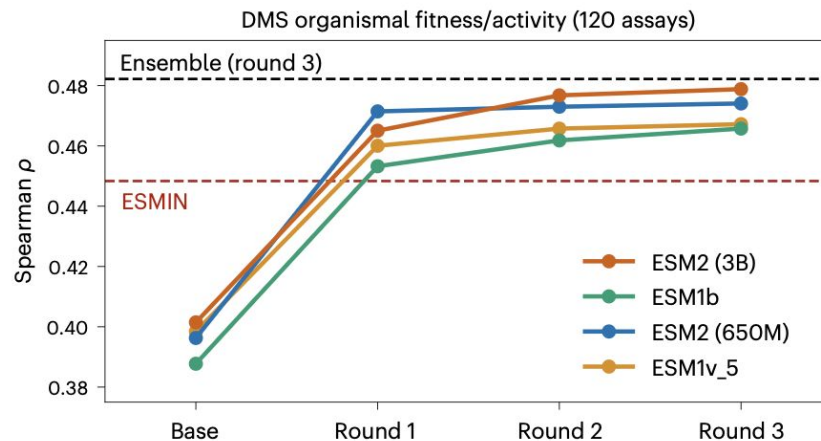
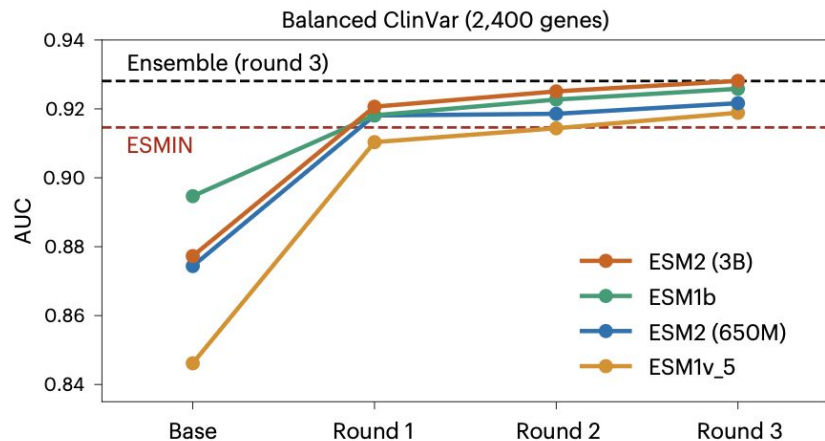
Co-distilled models are improved, with additional ensembling on both Balanced ClinVar and ProteinGym DMS benchmarks.



# VESM (3B)

c

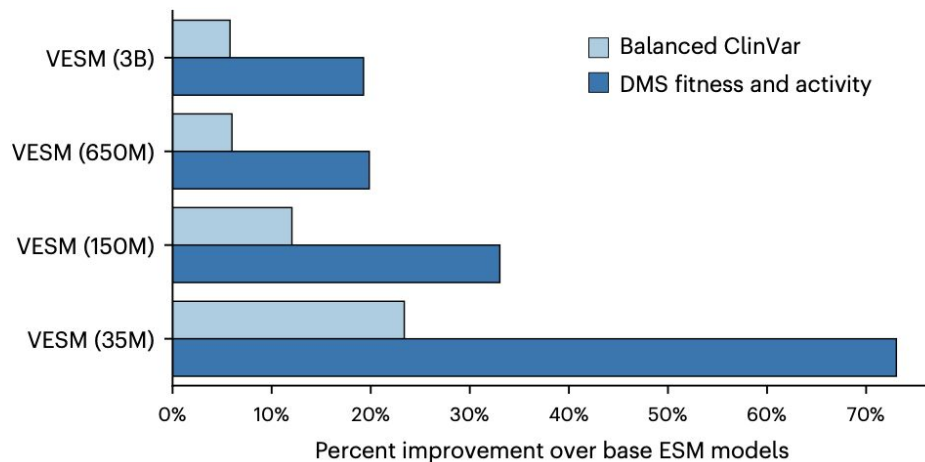
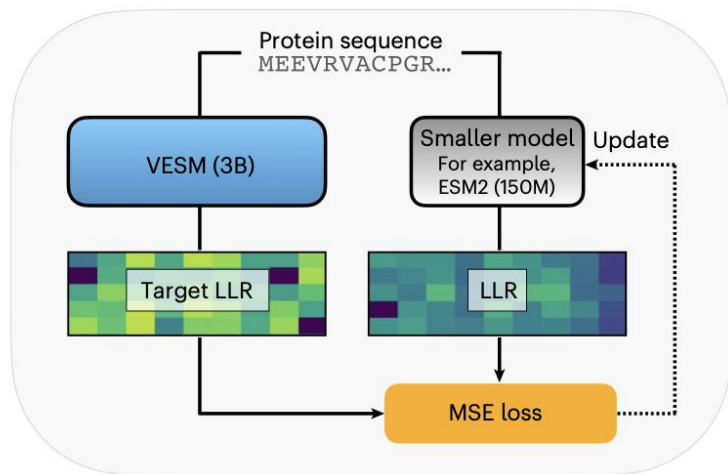
ESM2 (3B) converges to the ensemble's performance after multiple rounds of co-distillation



# LLR distillation of VESM-3B into smaller models

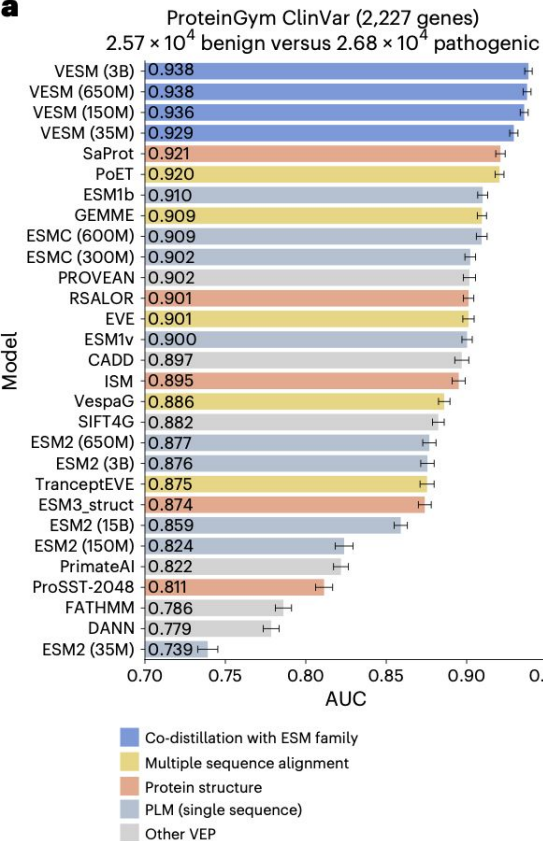
d

LLR distillation of the final round ESM2 (3B) into 650M, 150M and 35M parameter models

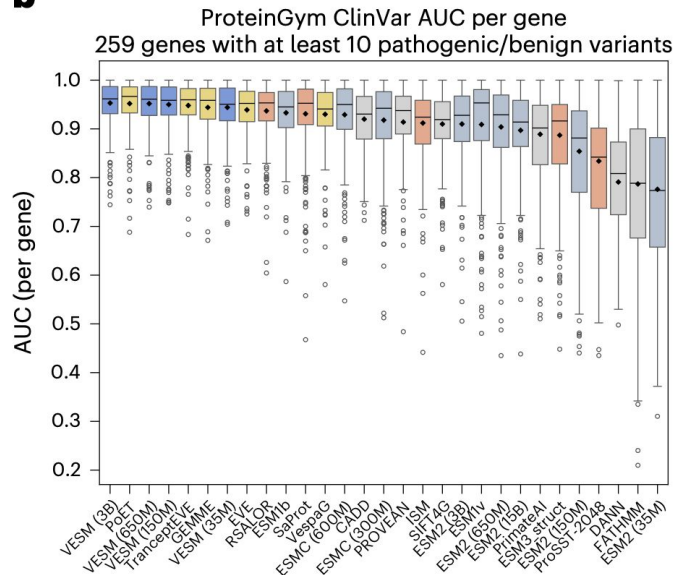


# Evaluation of sequence-only VESM models on clinical VEP

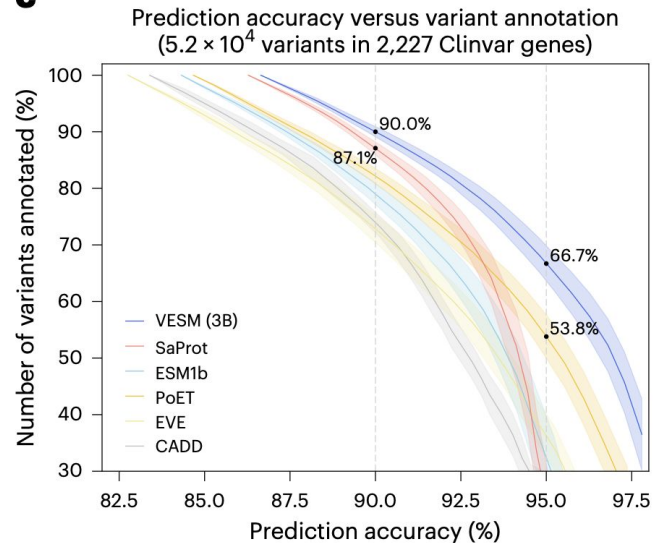
**a**



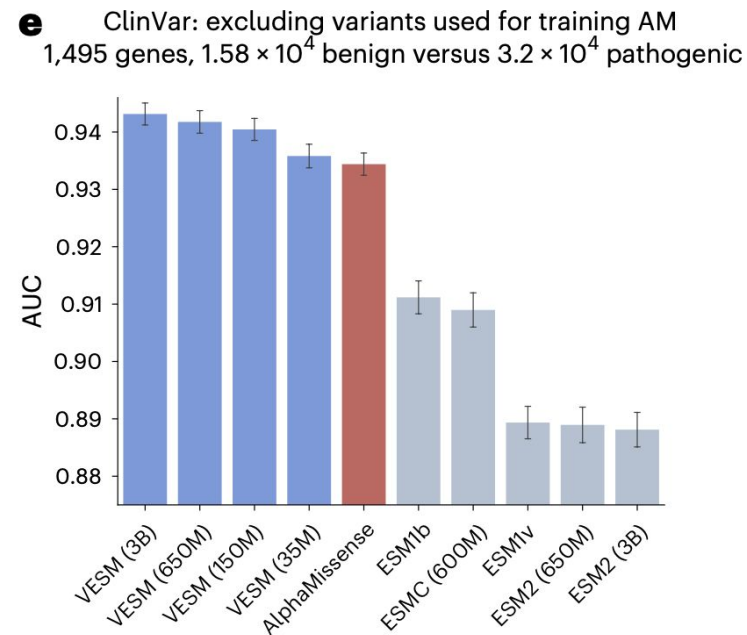
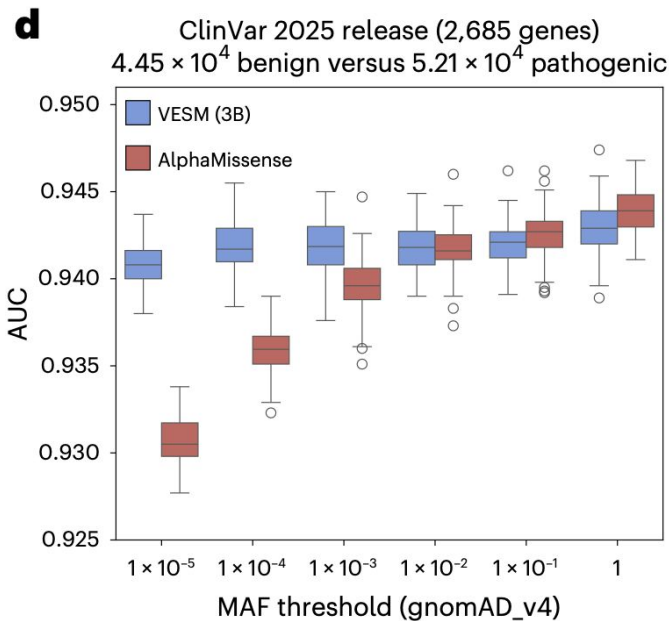
**b**



**c**



# VESM vs AlphaMissense



**f**

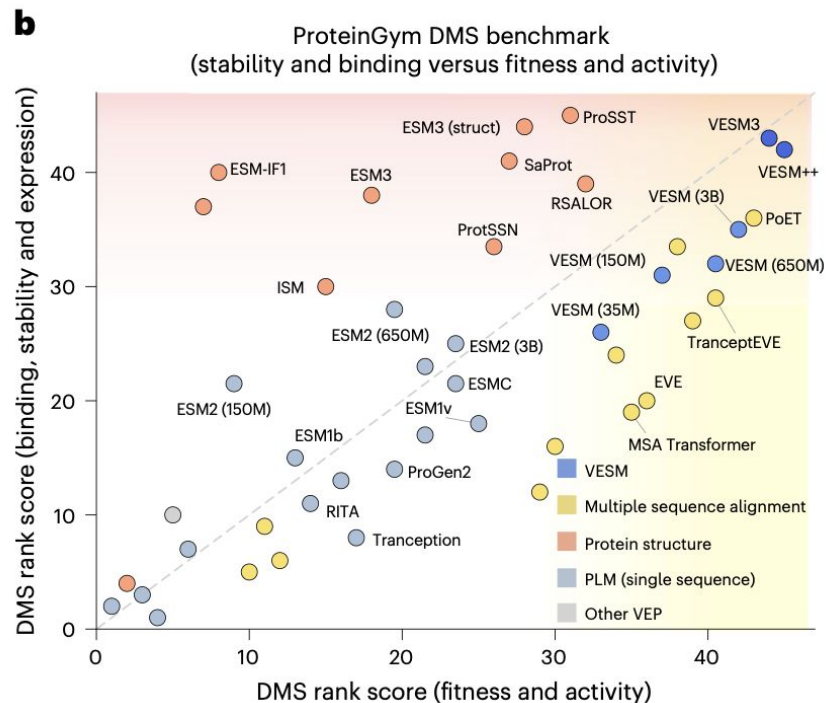
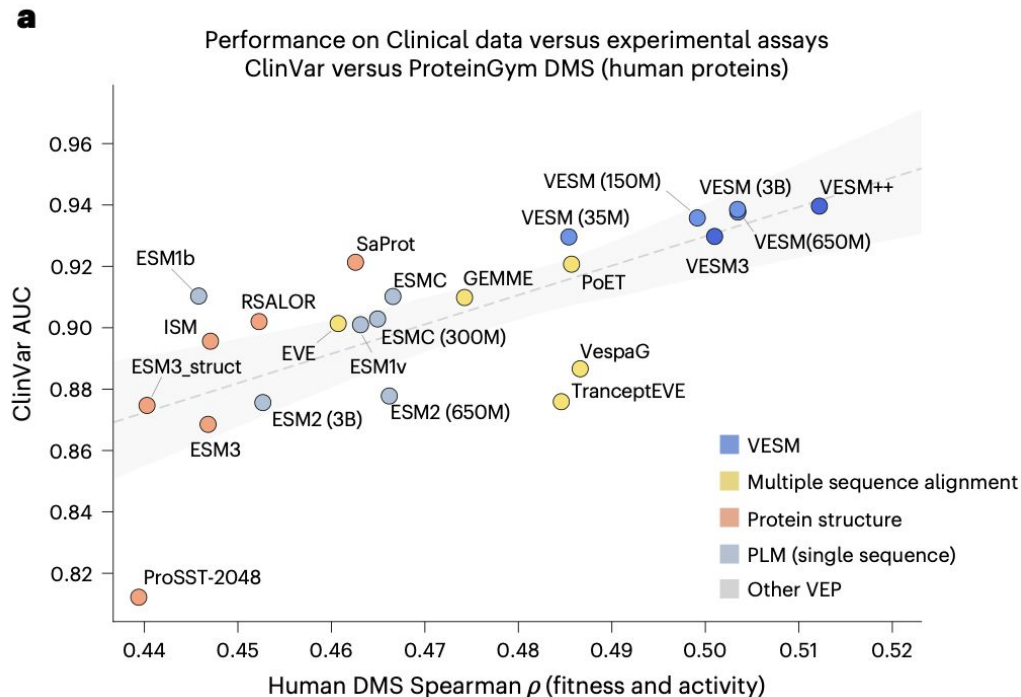
ClinVar, calibrated binary classification metrics (threshold of 0.5)

|               | Accuracy | Sensitivity | Specificity | Precision | DOR  | MCC   | F1 Score | Relative scale (per metric) |
|---------------|----------|-------------|-------------|-----------|------|-------|----------|-----------------------------|
| VESM (3B)     | 0.874    | 0.873       | 0.877       | 0.937     | 49.1 | 0.727 | 0.904    | 1.0                         |
| VESM (650M)   | 0.873    | 0.870       | 0.878       | 0.937     | 48.5 | 0.725 | 0.903    | 0.9                         |
| VESM (150M)   | 0.870    | 0.868       | 0.875       | 0.936     | 46.0 | 0.719 | 0.900    | 0.8                         |
| AlphaMissense | 0.866    | 0.869       | 0.858       | 0.928     | 40.3 | 0.707 | 0.898    | 0.7                         |
| VESM (35M)    | 0.862    | 0.858       | 0.871       | 0.933     | 40.9 | 0.703 | 0.894    | 0.6                         |
| ESM1b         | 0.851    | 0.857       | 0.839       | 0.917     | 31.1 | 0.674 | 0.886    | 0.5                         |
| ESMC (600M)   | 0.839    | 0.822       | 0.875       | 0.932     | 32.3 | 0.665 | 0.874    | 0.4                         |
| ESM2 (3B)     | 0.818    | 0.817       | 0.819       | 0.904     | 20.2 | 0.610 | 0.858    | 0.2                         |

# VESM++: Structure-aware ensemble model

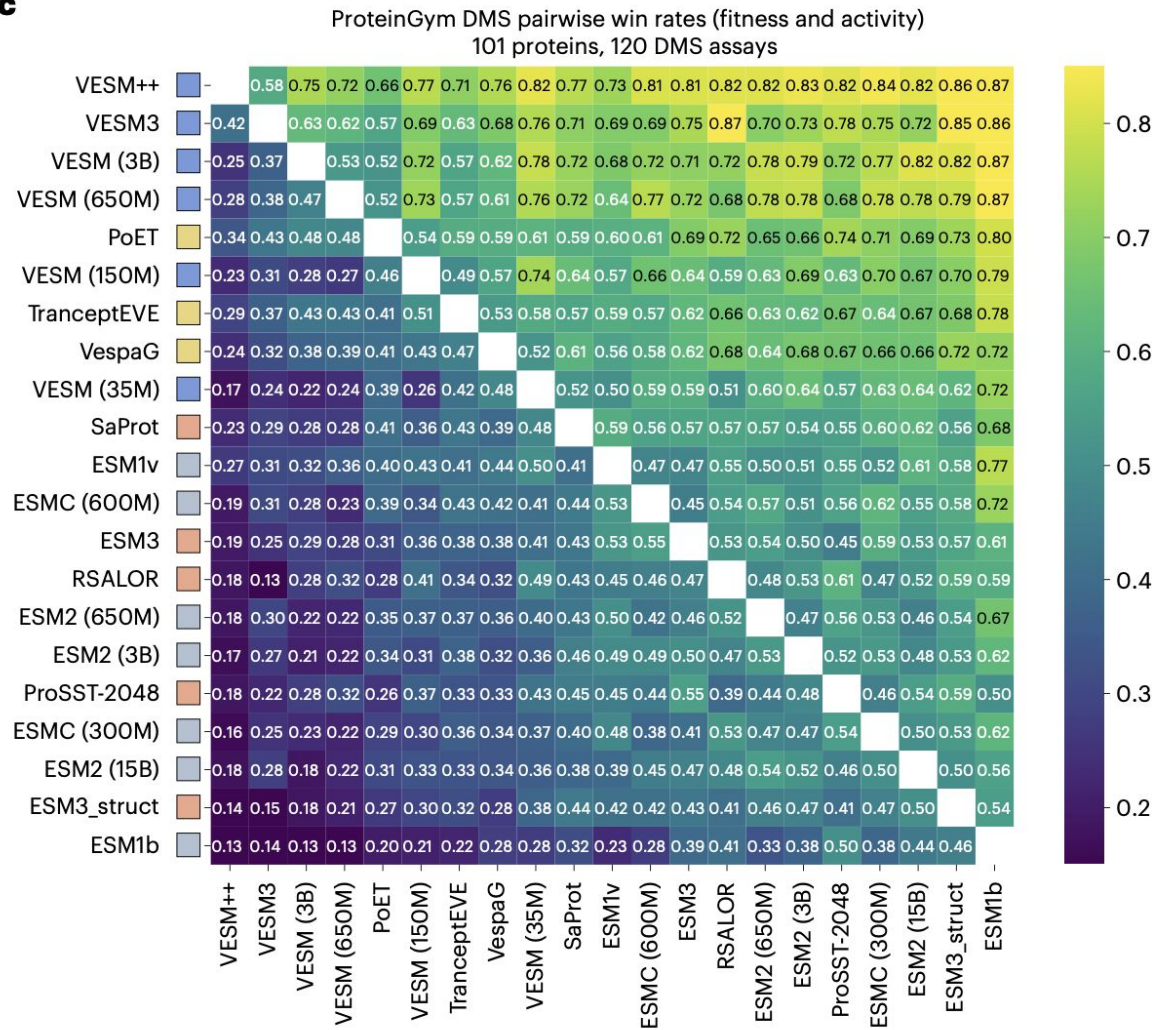
- Sequence and structure provide incomplete but complementary views of the protein fitness landscape.
- Naively combining sequence + structure does not guarantee better performance
- Structure-aware models may underutilize sequence information
- VESM → enhances sequence representations → plugged into structure model (ESM3) → VESM3
  - Fine-tune **only sequence components**
  - Use **sequence-based loss (MSE)**
  - Keep structure module intact
- VESM++ = VESM3 + VESM (3B)

# VESM achieves state-of-the-art VEP performance in both clinical and DMS benchmarks

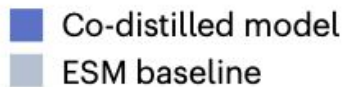


VESM++ and VESM3 models achieve the highest win rates overall, outperforming all MSA- and structure-based models. Even smaller sequence-only VESM models (except for 150M, 35M) outperform structure-based models.

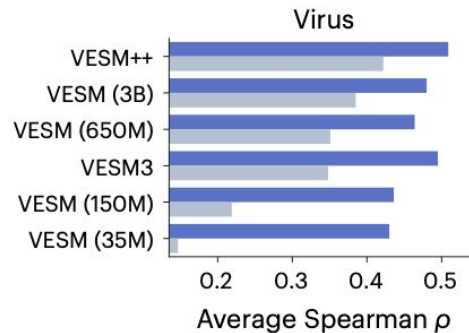
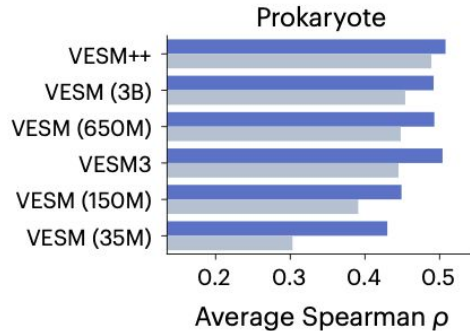
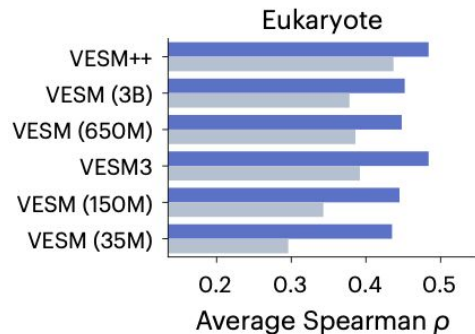
c



Model performance improved across taxonomic categories, with viral proteins showing the largest gains relative to other nonhuman proteins.

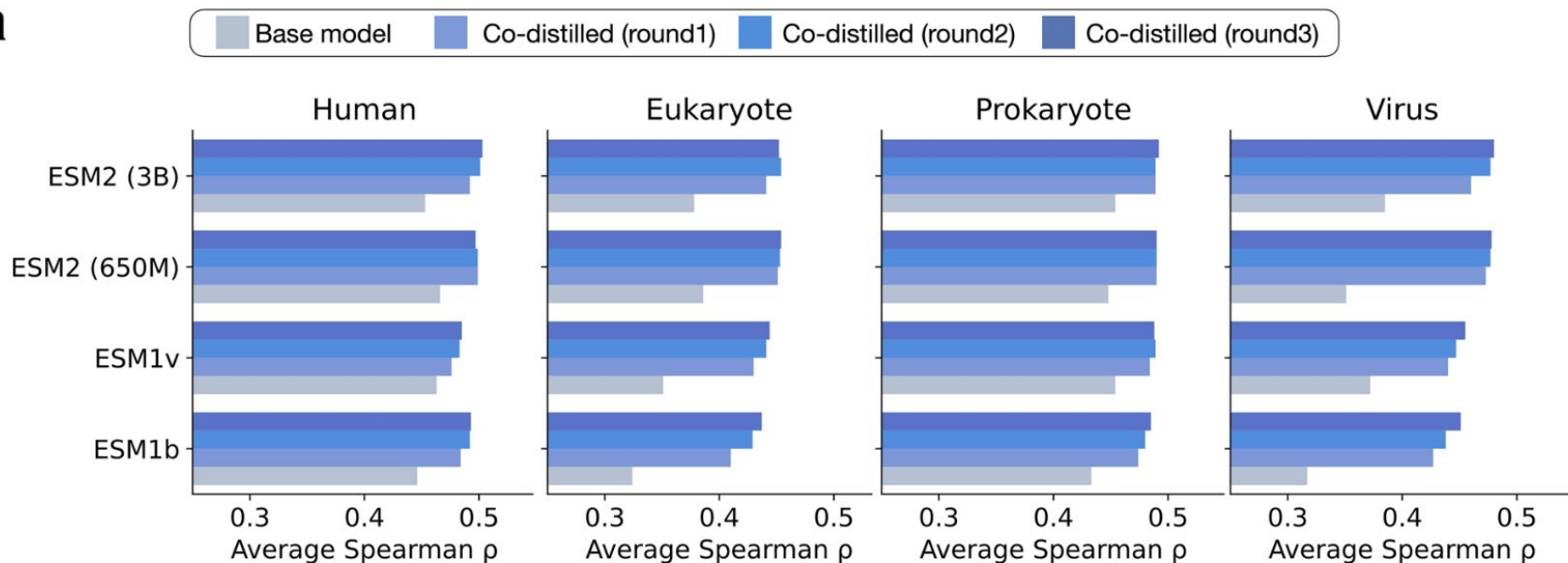


**d** Comparison to ESM base models  
(58 nonhuman proteins, 70 assays)



# Cross-taxon gains from co-distillation

**a**



# Linking Variant Scores to Clinical Signals

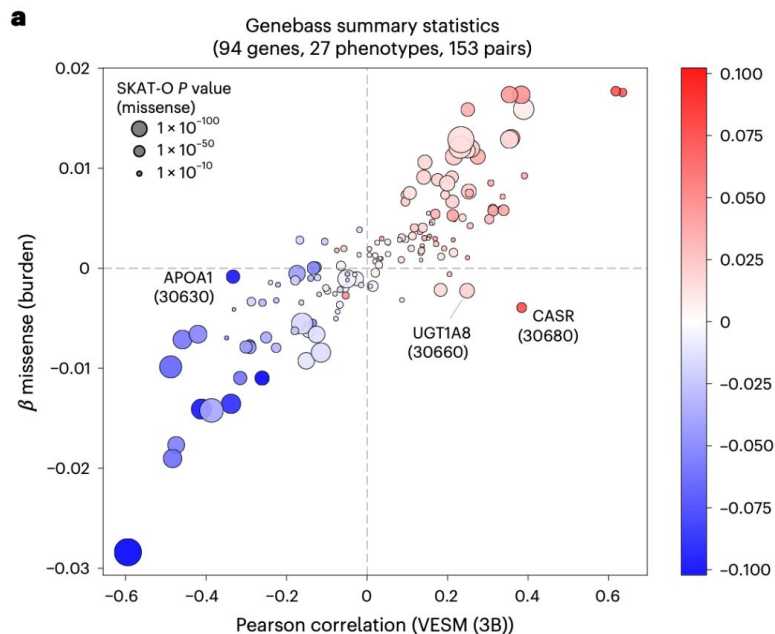
## VESM Captures Clinically Relevant Variant Effects

VESM-3B correlations align with:

- Gene-level missense burden effect sizes
- SKAT-O p-values

Model captures:

- Pathogenicity
- Effect magnitude
- Direction (risk vs protective)



# VESM Outperforms and Generalizes Across Traits

- Gene-trait association strength ( $-\log_{10}$  p-value)

VESM variants rank at the top:

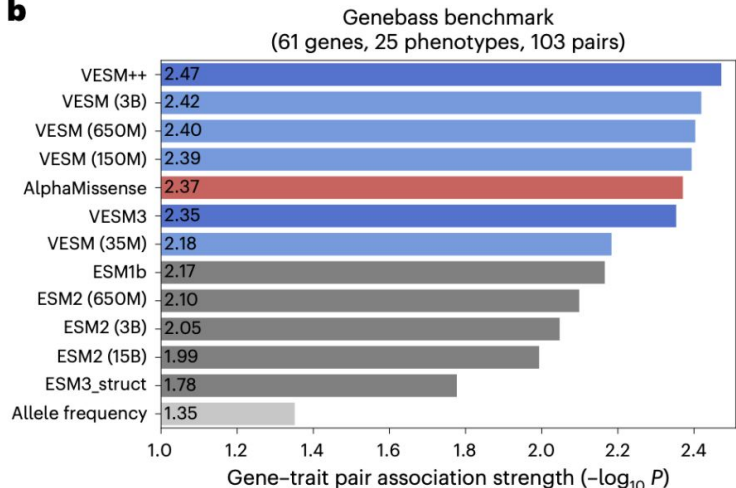
VESM++ / VESM (3B) outperform AlphaMissense, ESM2, ESM1b

Stronger ability to detect gene-phenotype links

- Performance across domains:
  - Lipid metabolism
  - Liver function
  - Renal function

VESM consistently strong across all categories

**b**

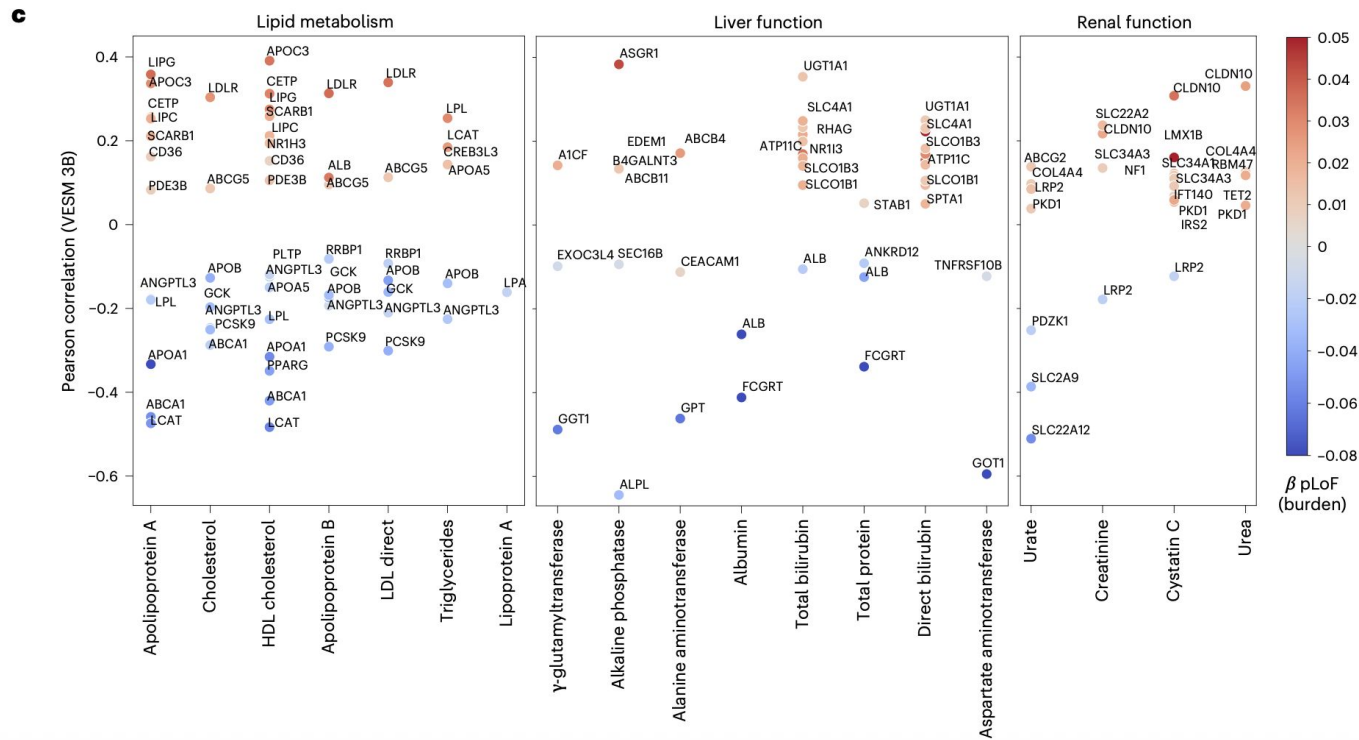


# ESM correlations align with pLoF signals

pLoF (Predicted Loss-of-Function):  
Variants that disrupt protein function  
(e.g., stop-gain, splice)

pLoF variants = strongest known functional signals

VESM captures missense effects at comparable signal strength



# Discussion

- Sequence-only PLMs can match or outperform hybrid models by leveraging complementarity within the ESM family.
- Co-distillation with maximum-confidence amplifies evolutionary signals and outperforms averaging
- Consistent gains across models and robustness to data size and homology
- VESM compresses this into a single efficient model that outperforms MSA-, structure-, and hybrid-based methods while generalizing across domains.
- Its predictions align with clinical, DMS, and pLoF signals, capturing both effect magnitude and direction.